OT PCAP Analysis

Purpose

Understanding the protocols that exist on OT networks is an important piece of trying to secure the systems. In the OT environment, network attacks have resulted in kinetic and operational consequences. Wireshark is an invaluable tool for understanding network traffic. This lab will ask students to dig into a set of packet capture (PCAP) files and answer a series of questions related to those PCAP files.

Instructions

Download Wireshark from https://www.wireshark.org. It is available for multiple operating systems, so you should be able to run it natively on whatever system you have.

Download the "`OT_PCAPs.zip`". https://github.com/jimgilsinn/BSidesDE2026/blob/main/OT_PCAPs.zip

OT_PCAP_1.pcap

1. What are the IP addresses involved in the conversation?
2. Which IP address is requesting information?
3. Which IP address is responding with the information?
4. What protocol is being used for this conversation?
5. What network port is being used for this conversation?
6. What data is being returned?
7. What is the approximate frequency that the data is being polled?

OT_PCAP_2.pcap

8. What are the IP addresses involved in the conversation?
9. What protocol is being used for this conversation?
10. What network port is being used for this conversation?
11. What is the model name?
12. What is the firmware revision?

OT_PCAP_3.pcap

13. What are the IP addresses involved in the conversation?
14. What protocol is being used for this conversation?
15. What network port is being used for this conversation?
16. What function code is used to initiate a Cold Restart?
17. What function code is used to initiate a Warm Restart?

OT_PCAP_4.pcap

18. What IP address sent the EtherNet/IP ListIdentity request?
19. What IP address sent the EtherNet/IP ListIdentity response?
20. What is the Vendor ID in the ListIdentity response?
21. What is the Product Name in the ListIdentity response?
22. What is the Revision in the ListIdentity response?

OT_PCAP_5.pcap

23. What IP address is requesting to update its time?
24. What is the DNS name of the server where it is requesting its time?
25. What is the IP address of the NTP server?
26. What is the Transmit Timestamp that the requester sends along with the NTP request?
27. What is the Root Delay in the NTP response from the server?

OT_PCAP_6.pcap

28. What are the IP addresses involved in this conversation?
29. What protocol is being used for this conversation?
30. What network port is being used for this conversation?
31. What is the Requested Packet Interval (RPI) in the CIP Connection Manager Forward Open packet?
32. What is the Originator Vendor ID in the CIP Connection Manager Forward Open Packet?
33. What is the T->O Connection ID?
34. What is the T->O data?
35. What is the O->T Connection ID?
36. What is the O->T data?